

Legacy of Equality: Black-White Teacher Pay Gap and Long-Term Mortality Among Black Students*

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Abstract

Despite a large literature linking education to long-term health and longevity, there remains limited causal evidence on how improvements in school quality shape later-life outcomes. During the early to mid-20th century, Black teachers earned lower salaries than their similarly qualified White counterparts due to de jure segregated pay schedules. In response, the National Association for the Advancement of Colored People (NAACP) embarked on a campaign to equalize teacher salaries through litigation and court rulings. In several Southern states, this litigation resulted in state policy reforms that restructured teacher pay schedules in an effort to increase Black teachers' salaries. Prior research has demonstrated that these reforms significantly increased Black teachers' salaries, by nearly 50%, and improved Black students' enrollment and grade progression. In this paper, we contribute to this literature by investigating the long-term impacts of these policies on students' longevity. We find that exposure to the reform during school age increases Black individuals' restricted life expectancy by approximately 2.2 additional months between ages 65 and 75. Through a battery of tests, we rule out concerns that our results are confounded by within-state cross-cohort changes in longevity, within-state race-specific improvements in overall longevity, or socioeconomic and sociodemographic compositional changes between treated and control cohorts. We discuss the policy implications of these findings in light of the literature on the later-life impacts of schooling inputs, emphasizing the imperative to address racial disparities in educational access and quality, which play a crucial role in shaping health outcomes among Black individuals.

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1. Introduction

Persistent racial disparities in health and longevity today reflect deep-seated structural inequalities, with unequal access to quality education being a central factor. Historically, Black schools in the Jim Crow South were chronically under-resourced relative to White schools, a pattern that helped perpetuate broader socioeconomic gaps. Economists have documented a powerful “education-health gradient”: more schooling substantially improves adult health and survival. For example, studies find that four additional years of education can reduce five-year mortality rates by roughly 1.8 percentage points, and that the health return to education is comparable in magnitude to major demographic factors (Cutler & Lleras-Muney, 2006). These links suggest that education has been a key channel through which historical inequities have translated into lifetime health disparities (Bailey et al., 2017; Gee & Ford, 2011).

A growing body of literature shows that children’s schooling experiences, especially teacher and school quality, shape adult socioeconomic status (SES) and health. High-value-added teachers raise students’ future earnings and SES: students assigned to high-value-added teachers are more likely to attend college, earn higher incomes, and live in higher-SES neighborhoods (Chetty et al., 2014). At the same time, better education is strongly associated with longer life. For instance, one study of Jim Crow-era Rosenwald schools finds that full childhood exposure to these schools (which increased Black educational attainment) raised later life expectancy by 2–3 months (Aaronson et al., 2021). In short, improved school quality can have lasting economic and health benefits.

Despite this, little empirical work has evaluated targeted reforms to segregated school quality. We focus on one such case: the NAACP’s mid-20th century campaign to equalize Black and White teacher salaries in the South. Beginning in the late 1930s the NAACP brought court

cases forcing Southern states to raise Black teachers' pay toward parity (Cascio & Lewis, 2024). These lawsuits and related policies sharply increased Black teachers' salaries in affected states.

Prior research has documented that salary equalization reforms not only increased pay for Black teachers but also improved school enrollment and educational attainment among Black students (Cascio & Lewis, 2024). However, we know far less about whether these reforms had long-term effects on students' survival. We hypothesize that these educational improvements—driven by more equitable compensation for teachers—may have had lasting effects on health and survival among their students, potentially contributing to a narrowing of the Black-White longevity gap. Understanding this link is critical, as it sheds light on how investments in educational equity can have long term impacts on health. To test this hypothesis, we examine whether exposure during school ages to the NAACP's legal victories and the resulting increases in Black teacher salaries led to improvements in life expectancy among Black individuals. We also investigate potential mechanisms by assessing whether these legal victories improved educational attainment in the short term and socioeconomic status in the medium term, which may help explain the observed long-run effects on survival.

To provide one of the first studies examining the effects of Black teachers wage increase on Black students' long term longevity, we use restricted-access death records from the National Center for Health Statistics and exploit the staggered timing of the NAACP's first legal victory for teacher salary equalization. Our empirical strategy estimates event study models that compare mortality outcomes among Black individuals who were school-aged at the time of the reform to those who were older, relative to the same age-based differences among White individuals.

Our empirical strategy relies on the standard parallel trends assumption, which we assess using placebo tests embedded in our event study framework. Specifically, we examine mortality

trends among individuals who were 17 or older at the time of the NAACP's legal victory that led to increases in Black teacher salaries, cohorts that should have had no exposure to the policy change during schooling. If the assumption holds, we should observe no effects for these older cohorts. The effects then grow gradually as younger cohorts gain greater exposure (e.g., those aged 17, then 16, and so on at the time of the policy change). Consistent with this expectation, we find no significant trends among those 17 and older and observe a clear pattern of increasing effects with greater exposure among younger cohorts.

We find that Black individuals exposed to the reform during their schooling years gained approximately 2.2 additional months of life between ages 65 and 75, measured as restricted life expectancy on a census population base within a balanced observation window. This estimate is statistically significant and robust across a range of specifications. These gains in survival are likely mediated by improvements in educational quality and subsequent socioeconomic conditions. Consistent with this interpretation, we observe meaningful gains in educational attainment and socioeconomic status.

To further validate our identification strategy, we conduct a placebo analysis by examining the effects of NAACP legal victories on White individuals, a group unlikely to have been directly affected. If our main results capture the true causal impact of these reforms, rather than reflecting coincidental shocks or unobserved confounders, we would expect the estimated effects for White individuals to be minimal and statistically insignificant. Consistent with this expectation, we find little to no impact on White students, reinforcing the interpretation that our results are driven by the NAACP's legal efforts and not by other contemporaneous factors that have the same effects on both racial groups.

Our study makes three contributions to the literature. First, we contribute to research on racial disparities in school resources, with a specific focus on the Black-White teacher pay gap and its implications for Black student outcomes. While a broad literature links increased school funding to improved academic achievement and labor market success (Biasi, 2023; Brunner et al., 2019; Card & Krueger, 1992a, 1992b, 1996; Card & Payne, 2002; Jackson et al., 2016, 2021; Lafortune et al., 2018; Liscow, 2018; Rothstein & Schanzenbach, 2022; Springer et al., 2009), few studies isolate teacher compensation as a key mechanism. Our work builds on a narrower historical literature that documents how systemic underinvestment in Black schools under Jim Crow shaped long-run racial gaps in education and earnings (Carruthers & Wanamaker, 2017), and how targeted investments—such as the Rosenwald school program—yielded lasting gains in schooling, wages, migration, and even longevity (Aaronson et al., 2021). We extend this line of research by focusing on the effects of NAACP-driven teacher salary equalization, showing that more equitable pay for Black educators led to meaningful improvements in the health and longevity of Black students. In doing so, we highlight the long-run consequences of structural investment in educators and the critical role of teacher compensation in narrowing racial disparities.

Second, we contribute to the literature on early-life exposures and later-life mortality by highlighting how policy-driven improvements in school quality can influence long-term health, particularly for historically marginalized populations. While prior work links early-life conditions to mortality (Aizer et al., 2016; Almond et al., 2018; Duncan et al., 2025; Fletcher & Noghanibehambari, 2023, 2024; Hayward & Gorman, 2004; Montez & Hayward, 2014; Noghanibehambari et al., 2024; Noghanibehambari, Slusky, et al., 2025; Noghanibehambari, Vu, et al., 2025; Van Den Berg et al., 2006; Vu et al., 2023), few studies consider how racially targeted

reforms in education shape disparities in later-life health. Our findings suggest that correcting racial inequities in educational resources may yield long-term health returns.

Third, we add to the limited empirical literature on the long-run impacts of the NAACP's legal activism. While the organization's legal victories are well-documented qualitatively (Kirk, 2009; Meier & Bracey, 1993; Weinberger, 2011; Zangrando, 1965), few studies have quantitatively assessed their downstream effects on education and health (Cascio & Lewis, 2024; Donohue et al., 2002). We are among the first to link these legal efforts to gains in longevity among Black individuals, revealing the broader legacy of civil rights advocacy in reducing racial health disparities.

The rest of the paper is organized as follows. Section 2 reviews the background of the NAACP litigations and the resulting policy reforms. Section 3 discusses the data sources and sample selection procedure. Section 3.2 introduces the econometric method. Section 4 presents the results. Section 5 concludes.

2. Background

In the decades following the *Plessy v. Ferguson* ruling (1896), which upheld the legality of segregation, the NAACP embarked on a mission to address the unequal distribution of school resources between Black and White schools, with one of the primary goals being to increase the salaries of Black teachers. Black educators were paid significantly less than their White counterparts with similar qualifications. In the NAACP's view, this unequal pay schedule for Black educators was in violation of the separate but equal principle. To fight this inequality, the NAACP supported a series of court cases through the 1930s and 1940s. These lawsuits sought to challenge the unequal pay of Black educators in federal courts. However, the NAACP's support for these cases was driven by broader and deeper goals than merely achieving individual- or

district-level equality. By securing legal victories, the NAACP aimed to pressure other school districts into voluntarily adopting equal pay schedules and, ultimately, to establish a statewide mandate for teacher salary equity.

Not all the court cases were successful, and educators who filed lawsuits often faced retaliation from school districts. In March 1939, Aline Black, a Norfolk high school teacher, filed the first teacher salary equalization suit in Virginia. The school board declined to renew her contract in June 1939, and the courts held that she consequently lacked standing. The NAACP then brought suit on behalf of Melvin O. Alston. On June 18, 1940, the Fourth Circuit held in *Alston v. School Board of the City of Norfolk*, that paying Black teachers less than White teachers of equal qualification violated the Equal Protection Clause of the Fourteenth Amendment. The Supreme Court denied certiorari in November 1940, leaving the Fourth Circuit's judgment in place without itself ruling on the merits. Because a denial of certiorari establishes no Supreme Court precedent, *Alston* was binding only within the Fourth Circuit; in the remaining states in our sample it operated as persuasive authority and as a template for subsequent NAACP litigation, each state entering our analysis at the date of its own first successful case.

In the face of such litigations, several states reinforced these court-mandated teacher salary equalizations by restructuring minimum teacher salary schedules and forcing districts to follow such schedules in order to receive state education aid (Cascio & Lewis, 2024). At the time, these schedules allowed for lower Black teacher salaries as they were explicitly divided by race. However, the new teacher salary schedules were solely based on training and experience. Such policy-induced salary schedules would promote greater equity in teacher compensation, narrowing racial inequality in resources, with potential impacts on students.

3. Data Sources

The primary data source utilized in this study is 1979-2020 restricted-access death records from the National Center for Health Statistics (NCHS, 2020). The NCHS data reports the universe of deaths in the U.S. with their underlying causes. It contains limited demographic information, such as race and gender, and report the deceased's state of birth and age at death. The latter two variables allow us to infer the individual's birth year and place, which is crucial to our analysis. A potential concern is that individuals may have attended school in a different state than where they were born, leading to mismeasurement of exposure to NAACP litigation. We will address this issue later in the paper.

We derived the main estimation sample using the following criteria. First, we restricted the sample to individuals identified as Black or White. Second, we restricted the sample to birth cohorts of 1915-1945. This window allows us to observe multiple cohorts born before and after the NAACP's first legal victory in teacher pay equalization. Third, we limited the sample to cohorts that were born in states with at least one successful NAACP Teacher Salary Equalization Case by 1948. These states are Alabama, Arkansas, Georgia, Kentucky, Louisiana, Maryland, Tennessee, Texas, and Virginia.⁶ Finally, we restricted attention to deaths occurring between ages 65 and 74. Because the mortality data cover 1979 through 2020, this band is observed over an identical age range for every cohort in the sample, including those born in the 1940s. This balance is what makes the measure comparable across cohorts, and it focuses the analysis on old-age mortality rather than on deaths that may have occurred earlier due to unrelated causes.

⁶ We excluded Florida and South Carolina because prior research indicates that both states implemented changes to teacher minimum salaries that, while seemingly equitable, ultimately perpetuated discrimination (Cascio & Lewis, 2024). However, (Margo, 1990, Table 4.1) shows that by 1950, the Black–White teacher pay gap in these states was comparable to that in other Southern states. Our results remain robust when Florida and South Carolina are included in the sample.

The second data source used in this study is the information on the NAACP Teacher Salary Equalization Cases by state which we obtained from Murray (1947) and Cascio & Lewis (2024). These sources provide information on the timing of the first successful NAACP cases in Former Confederacy states and border states. To merge this data with the NCHS, we match it based on the individual's state of birth and year of birth. Figure 1 shows the statewide pattern in the year of the first successful NAACP litigation.

Table 1 reports summary statistics for the analysis sample. The data consist of 450 birth-state by birth-year by race cells, covering nine birth states and 29 birth cohorts born between 1915 and 1945. The sample is restricted to states that had at least one successful NAACP teacher salary equalization case by 1948. On average, 72% of cells were exposed to NAACP teacher salary equalization, with individuals exposed for an average of approximately 6.1 years during school age. The mean restricted life expectancy between ages 65 and 75 is 104.7 months, and 88.9% and 75.4% of individuals survive to ages 70 and 75, respectively. These averages mask substantial racial disparities. Black cells have lower restricted life expectancy (102.2 versus 105.4 months), lower survival to age 70 (86.8% versus 89.5%), and lower survival to age 75 (72.2% versus 76.2%) than White cells.

A difference of 3.2 months of restricted life expectancy may appear small relative to the 120-month window,⁷ but the window itself bounds how large any difference can be: because everyone who survives to age 75 is assigned the full 120 months regardless of how much longer they live, the measure compresses longevity differences by construction. Relative to the mortality burden actually borne in the window, White cohorts lose 14.6 (=120-105.4) of the possible 120

⁷ The measure is bounded at 120 months (10 years $\times$ 12 months).

months to death between ages 65 and 75, while Black cohorts lose 17.8 months ($=120-102.2$), 22 percent more.

In our regression analysis, we also incorporate birth-state-by-birth-year controls as covariates. These covariates are derived from the full-count decennial censuses from 1910 to 1940 and linearly interpolated for the inter-decennial years (Ruggles et al., 2020). They include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score.

3.1. Measuring longevity within a balanced observation window

Our mortality data record deaths occurring between 1979 and 2020. Because we observe only deaths inside this calendar window, each birth cohort is observed over a different range of ages: for example, the 1915 cohort is observed dying between ages 64 and 105, the 1930 cohort between 49 and 90, and the 1945 cohort between 34 and 75.

Birth cohort	Age in 1979	Age in 2020	Observable age range
1915	64	105	64–105
1930	49	90	49–90
1945	34	75	34–75

This has an important implication for any outcome measured as age at death. Older cohorts can be observed dying at very old ages but never young (their earlier deaths predate 1979), while younger cohorts can be observed dying young but never old (their later deaths occur after 2020). The observed mean age at death is therefore mechanically related to birth year, and because exposure to the reform is itself a function of birth year, this relationship can generate a spurious event-study pattern even absent any true effect.

To eliminate this bias we compare cohorts only over an age range that every cohort is observed to pass through. The ages at which all cohorts born 1915–1945 are observed within 1979–

2020 are 65 through 75. We therefore study mortality between ages 65 and 75, a “balanced” window in which every cohort is followed over the identical age interval, retaining the full set of birth cohorts and the full range of reform exposure.

Within this window we do not use the mean age at death among those who die between 65 and 75, because whether a person dies in this window is itself affected by the reform: if the reform reduces mortality, some individuals who would have died in their early seventies instead survive past 75 and leave the window, so the conditional mean is a selected quantity. Instead we use the restricted life expectancy between 65 and 75, the expected years of life lived in that interval among those alive at 65, equivalently the area under the survival curve (Royston & Parmar, 2013). This measure is known in the biostatistical literature as the restricted mean survival time and in demography as partial or temporary life expectancy; it is widely used to compare mortality across groups when the observation window must be bounded (Knop et al., 2023; Sasson, 2016; Valverde et al., 2021). A person who dies at 70 contributes five years; a person who survives to 75 contributes the full ten. We construct it at the birth-state by birth-year by race cell level, with number of deaths in the numerator and the census population alive at 65 in the denominator. The treatment effect is directly interpretable as additional months of life lived between 65 and 75.

In Appendix A, we compare this measure with the alternative of using mean age at death directly among individuals who die between ages 65 and 75. Figure A-1 illustrates the performance of the two candidate outcomes within this balanced age window. In the upper row, where the true treatment effect is zero, both restricted life expectancy and the mean age at death among individuals who die within the window are correctly flat. In the lower row, we impose a genuine reduction in mortality. Restricted life expectancy closely tracks the true effect, whereas the mean age at death recovers only about one-seventh of the effect. The reason is that individuals whose

deaths are averted survive beyond age 75 and therefore leave the 65–75 window. As a result, conditioning on death within this window conditions on an outcome that is itself affected by the treatment.

3.2. Empirical Strategy

Our empirical strategy leverages state-level variation in the timing of the NAACP’s first legal victory that led to improved pay for Black teachers, generating individual-level differences in exposure based on age at the time of the reform. Specifically, we estimate difference-in-differences models that compare changes in longevity among Black individuals who were of school age during the reform to those who were older, relative to the same age-based differences among White individuals. Specifically, we estimated models of the following form:

$$Y_{rbt} = \alpha + \sum_{j=-6}^{18} \lambda_j I(j = t + 17 - T_b^*) \times Black_r + \delta X_{rbt} + \theta Z_{bt} + \zeta_{bt} + \varepsilon_{rbt} \quad (1)$$

Where Y_{lbt} is the restricted life expectancy between ages 65 and 75, the expected months of life lived in that interval among those alive at 65, measured at the birth-state $\times$ birth-year $\times$ race cell level on a census population base and weighted by the population alive at 65 for race group r in birth state b and birth cohort t .

The function $I(.)$ generates dummy variables indicating the distance (in years) between the year a cohort turned 17 and the state-specific year of the first successful NAACP case. Negative event-time coefficients correspond to cohorts who turned 18 or older before the case and are thus arguably unexposed to the reform. In contrast, positive event-time coefficients represent individuals who were partially or fully exposed to the post-NAACP era during their school years. For example, a positive event-time coefficient of 10 suggests that the cohort experienced 10 years

of exposure to the reform during school age. $Black_i$ is an indicator variable that takes on a value of 1 if the person is Black and 0 if the person is White.

The main parameters of interest are the event study coefficients, λ_j . T_b^* indicates the year in which birth state b experienced a successful NAACP litigation. Each λ_j therefore capture the difference-in-differences estimate that track the effects of exposure to the NAACP litigation for Black individuals with j year of exposure relative to age 17, relative to the same age difference among White individuals.

The matrix X_{ibst} includes controls for indicators for race and the female share of each birth-state by birth-year cell. Z_{bt} is birth-state-by-birth-year controls which include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score. Our preferred specification includes birth state-by-birth year fixed effects (ζ_{bt}). As a result, the primary source of variation comes from comparing differences in longevity between Black and White individuals (first difference) across years after versus before the reform (second difference). It is reasonable to assume that increases in Black teacher salaries did not spill over to White students. Given the strict racial segregation laws in the American South during the 1940s and 1950s, it was extremely rare for Black teachers to teach White students. These systems enforced strict racial segregation, with Black teachers typically employed in Black schools and White teachers in White schools (Tischauser, 2012).

The inclusion of birth state-by-birth year fixed effects also helps mitigate concerns about other state-level policy shocks or changes in sociodemographic and socioeconomic conditions that could otherwise bias our estimates. Finally, ε is a disturbance term. Standard errors are clustered at the birth state by birth year cell level, which allows for correlation between the Black and White

cells of the same birth state and birth cohort. Table 4 reports an alternative in which standard errors are clustered on birth state.

While the event study coefficients offer a clear view of how the effects evolve across cohorts, we also estimate average treatment effects using the following regression specification:

$$Y_{rbt} = \alpha + \beta_1 Exposure_{bt}^{5-17} \times Black_r + \beta_2 Exposure_{bt}^{5-17} + \delta X_{rbt} + \theta Z_{bt} + \zeta_t + \gamma_b + \varepsilon_{rbt} \quad (2)$$

Exposure is a dummy variable indicating whether NAACP litigation in the birth state occurred at any time during ages five to 17—that is, whether the individual was exposed to the post-reform era at any point during their school-age years. In these regressions, we avoid interacting birth state-by-birth year fixed effects in order to identify the main effect of the exposure variable. The coefficient on this variable (β_2) represents the effect of NAACP litigation on White individuals, serving as a placebo check. The coefficient of interest β_1 captures the effect of exposure to the reform on Black individuals who were of school age during the reform to those who were older, relative to the same age-based differences among White individuals.

In this specification, we are able to identify and interpret the main effect of the exposure variable for White individuals. This coefficient, β_2 , captures overall state-level temporal changes in educational policies, spending, or other factors that may influence student outcomes and correlate with the timing of NAACP litigation. The estimated coefficient of *Exposure* in Table 2 is relatively small in magnitude, suggesting that such broader state-level trends are unlikely to be an important driver of our main results.

Our preferred specification, however, reveals the differential effect of exposure on Black (versus White) individuals by interacting the exposure variable with a Black indicator and

including birth state-by-birth year fixed effects. These fixed effects allow us to compare outcomes between Black and White individuals within the same birth state and cohort,⁸ while also absorbing any unobserved, cohort-specific factors at the state level—such as shifts in demographics, school funding, or policy changes unrelated to the NAACP litigation. This tighter control provides a cleaner identification of β_1 , our primary parameter of interest. Therefore, our preferred specification is as follows:

$$Y_{ibst} = \alpha + \beta_1 \text{Exposure}_{bt}^{5-17} \times \text{Black}_i + \delta X_{ibst} + \zeta_{bt} + \xi_s + \varepsilon_{ibst} \quad (3)$$

All parameters and covariates are similar to equation (1).

4. Results

4.1. Identification Assumption

Our empirical strategy relies on the standard parallel trends assumption. In our setting, this implies that the timing of the first successful NAACP litigation—and, by extension, individuals' ages at the time of that legal victory—were conditionally unrelated to the counterfactual trends in mortality. This assumption would be violated if, for example, the NAACP targeted states where Black children's outcomes were already improving in the absence of litigation, or if the first successful case coincided with other state-level policies that independently influenced long-run

⁸ Some readers may be concerned that, with a fixed education budget, raising Black teacher salaries might have required cuts to other school resources, potentially affecting White students and making them less suitable as controls. We note that some states responded to NAACP litigation by raising overall school funding rather than reallocating resources away from White schools. For example, beginning in the 1940s, Alabama introduced a school equalization program funded by a dedicated 1% sales tax and equalized property tax assessments, which together boosted public education funds by roughly 15% by 1960 (Porter, 2025). In South Carolina, the legislature enacted a 3¢ sales tax to support a wave of “equalization schools,” part of a broader effort to delay desegregation while improving Black school facilities (Dobrasko, 2022).

While Cascio and Lewis (2024) note that meeting uniform salary mandates for Black teachers could have required reallocating funds, they did not find any impact on White students' educational outcomes (Table 3, column 5). This suggests that funding for Black teacher pay increases did not come at the expense of White students.

outcomes. The evidence supporting this assumption comes from the placebo tests embedded in our event study design, particularly the absence of significant pre-trends beyond age 17—that is, insignificant estimates of λ_j for $j \in [-6, -1]$.

Further, we note that The NAACP pursued a deliberate and strategic approach when filing teacher salary equalization lawsuits, targeting localities where they believed they had the highest likelihood of success. Rather than spreading efforts evenly across all states with disparities, the NAACP chose jurisdictions with legal environments more conducive to civil rights litigation, such as areas with relatively progressive courts or where favorable legal precedents had already been established. They also sought strong, credible plaintiffs, often well-respected Black educators, and coordinated with local teachers’ associations to ensure community support (Kirk, 2009). This strategic targeting makes it less likely that the timing of teacher salary equalization cases was driven by time-varying factors influencing Black children’s long-term mortality, thereby lending additional credibility to the parallel trends assumption.

4.2. Threat to Identification: Secular improvements in Black longevity

A natural concern is that our estimates capture a general improvement in the longevity of Black Americans across birth cohorts rather than the effect of teacher salary equalization. The structure of our preferred specification narrows this concern substantially. Column 3 of Table 2, which we take as our benchmark and which is the specification used throughout Tables 3 and 4, absorbs interacted birth-state by birth-year fixed effects. Any factor that changed the mortality of Black and White cohorts born in the same state in the same year by the same amount is therefore differenced out, including improvements in medical technology, the expansion of Medicare and Medicaid, changes in nutrition and public health infrastructure, and state-level economic conditions. The identifying variation is not the level or trend of Black mortality, but the Black–

White gap within a birth state and birth cohort. For our estimates to be spurious, it is not sufficient that Black longevity was improving over this period; there must have been a differential trend in the racial gap, occurring within states, and timed to coincide with the state-specific arrival of NAACP litigation.

4.3. Main Results

Figure 2 presents the results of estimating Equation (1). The top panel displays OLS estimates, while the bottom panel uses the Sun and Abraham (2021) estimator to account for treatment effect heterogeneity across cohorts. The two approaches yield very similar patterns. Among southern African Americans, earlier exposure to teacher pay equity had positive effects on long term mortality. The estimates increase with years of school-age exposure: cohorts exposed to the reform for longer periods experienced progressively larger gains in longevity relative to individuals who were age 17 at the time of the reform. For example, individuals who turned 17 ten years after the reform, corresponding to ten years of exposure during school age, experienced a gain of roughly 1.5 months in life expectancy relative to those who were age 17 at the time of the reform. The estimated effect rises to more than three months for cohorts with the longest exposure. In contrast, we find no statistically significant effects for Black Americans who were ages 18 to 23 at the time of the reform (event times -1 through -6). These cohorts had little or no opportunity to benefit from the reform during school age, making the absence of detectable effects among them consistent with the proposed mechanism.

We do, however, observe some pre-trend in our event-study estimates. Although the pre-trend is not statistically significant, its magnitude can be read as sizable relative to our main estimate. We think a reasonable interpretation is that our data are not quite precise enough to distinguish between time trends and relatively small treatment effects, particularly effects in the

range of one to two months. Thus, the estimated effect may reflect a combination of underlying time trends and a relatively small treatment effect, which we believe is probably around one month. It is also useful to put this magnitude in context: the entire Black–White life-expectancy gap in our data is approximately 3.2 months. Thus, even an effect of one month would represent a sizable share of the overall Black–White gap.

Role of Staggered Treatment Timing.—Recent studies indicate that the varying effects of treatments across different cohorts, distinguished by their timing, can complicate the interpretation of difference-in-differences findings. In our context, we have five different treatment cohorts as illustrated in Figure 1. (Sun & Abraham, 2021) demonstrate that if each of these cohort exhibits different patterns of time-dependent treatment effects, difference-in-differences estimates may become contaminated by effects from other time periods. We demonstrate that this variation in treatment timing is not a concern in our context by using the alternative estimation method proposed in Sun and Abraham (2021). The result from this approach is reported in the bottom panel of **Error! Reference source not found.** We observe coefficients that are quite similar to those from our main OLS results shown in the top panel.

The results of the regressions presented in Equation (3) can be found in Table 2. In the first column, we present results with race and gender controls, birth year fixed effects, and birth state fixed effects. Column 2 adds birth-state by birth-year controls, and column 3 adds birth state-by-birth year fixed effects. According to the fully parametrized model in column 3, Black children who were exposed to NAACP litigation exhibit an average increase in restricted life expectancy between ages 65 and 75 of approximately 2.2 months.

To contextualize the magnitude of the observed effects of NAACP litigation on Black children longevity, we compare our findings to those from the existing literature on early-life

exposures. Aaronson et al., (2021), for example, study the long-term effects of early-life exposure to the Rosenwald schools, a major philanthropic initiative that supported the construction of over 5,000 schools for Black children in the rural South during the early 20th century. They find that full exposure increased life expectancy by roughly 2–3 months, an intent-to-treat estimate. Because Rosenwald schools reached only about one-third of rural Black students at their peak (Aaronson & Mazumder, 2011), the implied treatment-on-the-treated effect is about 6–9 months.

By comparison, our estimates in Figure 2 suggest a difference of up to three months in restricted life expectancy between ages 65 and 75 for fully treated to untreated cohorts. Our intent-to-treat estimates are likely to be close to treatment-on-the-treated effects because of the nature and broad implementation of the intervention. The NAACP-led teacher salary equalization was implemented as a blanket policy that raised salaries for Black teachers across affected states. Consequently, nearly all Black students in these states were likely exposed to better-resourced teachers, leaving relatively little scope for noncompliance or differential treatment intensity. Our estimates therefore provide a close approximation to the treatment-on-the-treated effect. Under this interpretation, the longevity gains associated with the NAACP-led policy are approximately 33% to 50% as large as the implied treatment-on-the-treated effects of Rosenwald schools.

Scaling the Effect by Extra Dollars Per Student.— Table 7 reports pre-reform school teacher salary gap for the nine states in our sample, drawn from Card and Krueger (1992b) and measured in the school years 1936 and 1938, the last two observations before the first successful case in our sample. We compute the implied increase in annual teacher salary spending per Black pupil as the Black–white salary difference divided by the Black pupil–teacher ratio. Expressed in 2025 dollars, this quantity averages \$212 per pupil per year and ranges from \$122 in Arkansas to \$299 in Louisiana.

This provides a natural way to express our estimates in terms of extra dollars per student. Taking our baseline estimate of 2.2 months against the average increase of \$212 per pupil per year implies roughly one additional month of restricted life expectancy per \$100 of annual per-pupil spending. This calculation attributes the entire estimated effect to the salary channel, it should therefore be read as an upper bound on the return to expenditure, since to the extent that equalization also operated through teacher retention, experience, or other margins, the return to dollars alone is smaller.

Valuing the Longevity Gains and the Cost of Equalization.—Another way to assess the magnitude of the estimated effects is by applying the Value of a Statistical Life (VSL). While estimates vary slightly across studies, several suggest a VSL of approximately \$10 million in 2017 dollars (Colmer, 2020; Kniesner & Viscusi, 2019). Among the deaths observed for these cohorts, the average age at death is roughly 891 months, implying that each additional month of life is valued at about \$11,223 (\$10 million / 891).

About 1.72 million Black individuals in our sample, 72 percent of the 2.36 million Black individuals alive at age 65, belong to exposed cohorts. Multiplying by the estimated gain of 2.2 months yields roughly 3.78 million life-months, equivalent to about 4,240 statistical lives (=315,333 life-years divided by 74.25 years, the 891-month mean age at death) and an aggregate benefit of approximately \$42.5 billion in 2017 dollars.

These benefits can be set against the cost of the policy. As reported in Table 7, full equalization implied additional teacher salary spending of about \$9.36 per Black pupil per year in contemporaneous dollars, or \$212 in 2025 dollars. Treated Black cohorts were exposed for an average of 8.5 school years, implying roughly 14.6 million pupil-years of additional spending and

a total cost of approximately \$3.1 billion in 2025 dollars. Undiscounted, the implied benefit–cost ratio is about 18 to 1.

This ratio is, however, sensitive to discounting, since costs are borne during childhood while benefits accrue nearly six decades later. Discounting benefits back to the mid-point of the school-age spending period, roughly 57 years before the benefits accrue, reduces the ratio to about 3 to 1 at a 3 percent discount rate and to approximately 1 to 1 at 5 percent.⁹ The cost figure is also conservative in one respect and generous in another: it excludes any administrative or litigation costs, but it counts only pupils who survived to age 65, whereas spending covered all enrolled children. We therefore regard these calculations as indicating that the intervention was plausibly cost-effective under moderate discounting, rather than as a precise estimate of its return.

Placebo test.— Researchers examining the long-term effects of the NAACP have found race-specific benefits concentrated solely on Black students and teachers (Cascio & Lewis, 2024). To further validate our identification strategy, we explored the effects of the NAACP on a group unlikely to be affected: White individuals. If our main estimates truly reflect the causal impact of the NAACP’s legal victories—rather than capturing unrelated, coincident shocks or unobserved interventions—we would expect the corresponding coefficients for White individuals to be small and statistically insignificant. Consistent with the race-specific nature of the intervention, the estimated effects for White individuals are substantially smaller in magnitude than those for Black individuals, although they are negative and statistically significant in Columns (1) and (2) (Row 2, Table 2). The contrast between the large positive effects for Black individuals and the much

⁹ Treated cohorts were exposed for an average of 8.5 school years ending at age 17, so we date costs to the midpoint of the spending period, age 13, and benefits to the midpoint of the 65-to-75 window, age 70, giving a horizon of 57 years. The corresponding discount factors are $1.03^{-57} = 0.186$ and $1.05^{-57} = 0.062$. Applied to the aggregate benefit of \$55.0 billion, these yield present values of \$10.2 billion at a 3 percent rate and \$3.4 billion at 5 percent. Costs of \$3.09 billion are incurred at the school-age reference date and are therefore not discounted. The implied benefit–cost ratios are 17.8 undiscounted, 3.3 at 3 percent, and 1.1 at 5 percent. All figures are in 2025 dollars.

smaller effects for White individuals provides additional support for the validity of our research design.

Heterogeneity by gender.— A key advantage of the large sample sizes in our data is the ability to generate reasonably precise estimates within sub-populations of interest. One notable factor through which effects may vary is gender, particularly since studies of educational programs often report differential effects for girls and boys. Table 3 shows the results of estimating our baseline specification separately for women and men. We find the effect is of similar magnitude for women and men. Specifically, the reform increased restricted life expectancy between ages 65 and 75 by 2.17 months for Black women (column 1) and 2.35 months for Black men (column 2), gains that are statistically indistinguishable across gender in this window. The similarity across gender suggests that the old-age survival benefits of improved school quality accrued broadly to both Black women and men.

4.4. Robustness Checks

Robustness to Mismeasurement of Exposure.— One potential concern is that individuals may have attended school in a different state from where they were born, and therefore exposure to the NAACP litigation would be mismeasured. A related concern is that migration itself may respond to the reform. Our outcome is constructed entirely by state of birth: both the deaths in the numerator and the census population in the denominator are attributed to the state in which an individual was born, so where individuals later reside does not mechanically affect the measure.

Figure 3 examines migration directly. The top panel reports an event study for the share of deaths at ages 65 to 75 that occur in the individual's state of birth. The coefficients are generally close to zero throughout the event window, and none is statistically distinguishable from zero. Thus, the estimates provide little evidence that exposure to NAACP teacher salary equalization

affected the likelihood that individuals died in their birth state. This result suggests that migration is unlikely to be an important source of bias in the main analysis.

The bottom panel of Figure 3 reports the corresponding event study for restricted life expectancy between ages 65 and 75 among non-migrants, where the denominator is the birth-state population scaled by the non-migrant share among decedents. The profile is flat before exposure and rises to approximately three months afterward, and the difference-in-differences estimate of 2.31 months (standard error = 0.53) is close to the full-sample estimate of 2.22 months. Because this imputation assumes that the non-migrant share among survivors equals the share among decedents, we regard the bottom-panel results as suggestive rather than as a formal robustness check.

Robustness Checks to Alternative Specifications.— Table 4 presents results from additional robustness checks using alternative specifications. Column 1 replicates the estimates from Table 2 (Column 3) and serves as the benchmark for comparison. Column 2 of Table 4 adds birth-state by race fixed effects, which allow the Black-White gap in restricted life expectancy to differ across birth states. The estimated effect remains similar at 1.938 months and is statistically significant at the 1 percent level.

Our main analysis assumes that teacher salary equalization in the states under study was implemented via statewide minimum teacher salary schedules. This assumption is supported by evidence indicating that such schedules served as the primary mechanism for equalization, were strong predictors of Black teacher salaries during the period (Card et al., 2022), and were mandated for districts to qualify for critical state education aid. However, not all states in our sample adopted statewide schedules. Specifically, Alabama, Louisiana, Tennessee, and Virginia implemented statewide minimum salary schedules, whereas other states relied on more localized approaches.

For example, Maryland’s equalization efforts were limited to county-level pay equity, Kentucky implemented equalization only at the district level, Arkansas relied on local equalization without a statewide schedule, and there is no evidence that Georgia and Texas adopted statewide minimum salary schedules during this period. Column 3 restricts the sample to the four states that implemented statewide minimum teacher salary schedules: Alabama, Louisiana, Tennessee, and Virginia. The estimate remains remarkably similar at 1.968 months, suggesting that the main findings are not driven by states in which equalization occurred primarily through more localized mechanisms.

Column 4 examines sensitivity to the functional form of the outcome by using the logarithm of restricted life expectancy between ages 65 and 75. The estimated coefficient of 0.022 implies an approximately 2.2 percent increase in restricted life expectancy, closely corresponding to the magnitude of the baseline estimate.

Columns 5 and 6 examine survival at specific ages as alternative outcome measures. Exposure increases the probability that Black individuals survive to age 70 by 1.671 percentage points and the probability of surviving to age 75 by 3.374 percentage points. Both estimates are statistically significant at the 1 percent level. The larger effect at age 75 is consistent with the improvement in restricted life expectancy observed in the main analysis and suggests that the benefits of exposure persist into later ages.

Finally, Column 7 clusters standard errors at the birth-state level rather than at the birth-state-by-year level. The estimated standard error is virtually identical to that in the benchmark specification. Overall, the results are highly stable across alternative fixed-effects structures, sample restrictions, functional forms, outcome measures, and approaches to statistical inference.

4.5. Mechanisms

Improved pay for African American teachers attracted more qualified individuals to the profession. With better compensation, these teachers were also more likely to invest in professional development, leading to higher-quality instruction for African American students. Research has shown that school and teacher quality have lasting impacts on students' adult socioeconomic status and long-term health (Aaronson et al. 2021; Chetty, Friedman, and Rockoff 2014). Socioeconomic status, in turn, is strongly linked to health and mortality outcomes. Individuals with higher socioeconomic status possess higher health-related knowledge and are less likely to engage in health risk behaviors, such as smoking and drinking (de Walque, 2007; Fletcher & Frisvold, 2011). They also experience better mental health, which can have downstream benefits for physical health and longevity (Cornaglia et al., 2015). Moreover, they are also more likely to have health insurance coverage, further supporting healthier and longer lives (Abaluck et al., 2021; Black et al., 2017).

Taken together, these pathways suggest that NAACP-driven litigation, by elevating teacher pay, improved the educational environment for Black students, boosting their later-life socioeconomic status and, ultimately, their longevity. To investigate these mediating channels, we analyze decennial census data from 1950–2000. We then apply the same sample selection criteria as in Section 3: (1) limit the sample to individuals identified as Black or White; (2) restrict to birth cohorts from 1915–1945; (3) include only cohorts born in states with at least one successful NAACP Teacher Salary Equalization case by 1948—Alabama, Arkansas, Georgia, Kentucky, Louisiana, Maryland, Tennessee, Texas, and Virginia; and (4) include only individuals who were aged 25 or older at the time of the Census in order to observe their completed high school education. We then estimate equation (1) to quantify how NAACP-litigation-induced policy changes affected Black students' short-term educational attainment—replicating the findings of

Cascio and Lewis (2024) using a different dataset—as well as on their medium-term occupational prestige.

Table 5 presents the results, with Panel A reporting estimates for women and Panel B for men. The interaction terms suggest improvements in educational and occupational measures for both Black women and men. In column 1 of Panel A, the estimated coefficient implies a 47 percent reduction ($0.014/0.030$) in the share of Black women with zero years of schooling, while the corresponding reduction for men is 50 percent in Panel B. These magnitudes are quite similar with the descriptive evidence from Cascio & Lewis (2024), who document a rise in Black first-to-second grade progression from 0.49 in 1939–40 to 0.84 in 1959–60 (a 71 percent increase relative to the mean).

Column 2 shows a 6.4 percentage point increase in the likelihood that a Black woman completed high school, representing a 14.5 percent gain relative to the mean (0.441). For men, the corresponding increase is 4.5 percentage points, or roughly 11.3 percent relative to the mean. Column 3's coefficient on the Occupational Prestige Index indicates a mid-career occupational prestige increase of about 16.5 percent for Black women ($4.45/27.04$) and 9.1 percent for Black men ($2.26/24.74$), suggesting that early exposure led to meaningful long-term socioeconomic gains.

Heterogeneity by cause of death.—

Table 6 explores the heterogeneity of the effects by cause of death. Examining cause-specific mortality allows us to assess further whether the longevity gains align with pathways that education and occupational outcomes was improved among Black students. The Cumulative Mortality Rates (CMR) are calculated using the following formula:

$$CMR_{bsrc} = \frac{Deaths_{bsrc}^{65-75}}{Population_{bsr}^{65}}$$

Here, b, s, r, and c represent the indices for birth cohort, birth state, race group, and cause of death, respectively. CMR is calculated for deaths occurring between ages 65 and 75. This age range is chosen because, within our observation window 1979–2020, individuals born in 1945 contribute deaths at ages 34–75, whereas those born in 1915 contribute deaths at ages 64–105. Restricting the analysis to deaths at ages 65–75 allows for a more balanced comparison across cohorts.

For the 1915 birth cohort, the denominator is the number of individuals alive in 1980 for each cohort–state–race–cause cell, using the 5% sample of the 1980 Census (Ruggles et al., 2020). For the 1925, 1935, and 1945 cohorts, we instead use the population counts from the 1990, 2000, and 2010 Censuses. This adjustment is necessary because, for example, the 1925 cohort only enters the death records at ages 65–75 beginning in 1990, making the 1980 population inappropriate for calculating their CMR.

For cohorts born between these census benchmark years, we assign the population from the year in which the cohort reaches age 65, using a population measure linearly interpolated between the two surrounding census years. For example, we use the 1981 population for the 1916 cohort, the 1982 population for the 1917 cohort, and so on up to the 2009 population for the 1944 cohort. Finally, we take the logarithm of the cumulative mortality rate (CMR).

Results in Table 6 show that exposure to teacher pay equalization reforms significantly reduced cause-specific mortality among Black individuals across all major categories. The largest reductions are observed for deaths from external and self-inflicted causes (column 5, –0.258), a category that includes many “deaths of despair,” such as those resulting from substance use and

suicide. This pattern aligns with prior research showing that financial losses and lower educational attainment are key factors contributing to deaths of despair, such as suicide and substance use (Fishman & Gutin, 2021). Note that this cause of death is not due to higher deaths of despair among White individuals, as the coefficient for Whites is small in magnitude and not statistically significant. Substantial declines are also seen for infectious diseases (column 1, -0.145).

The effects for chronic/non-communicable diseases (column 3) and other diseases (column 4) are smaller in magnitude. One possible explanation is that improvements in the detection and treatment of chronic conditions (cardiac care, oncology, diabetes management etc.) over the 1980s–2000s led to broad secular declines in chronic-disease mortality across the population. For instance, death rates for heart disease and stroke declined by approximately two-thirds and three-quarters, respectively, due to reductions in key risk factors, improved hypertension and cholesterol management, and advances in medical care (Centers for Disease Control and Prevention, 2011).

The main effects of “Exposure,” which capture the estimated impact of teacher pay equalization reforms on White individuals, are smaller in magnitude and statistically significant only for infectious diseases.

5. Conclusion

In conclusion, our study underscores the pivotal role of policy interventions, particularly those driven by the NAACP litigations and efforts to equalize teacher salaries, in mitigating Black-White disparities in health outcomes. While investments in school quality and teacher salaries may initially appear costly, they yield substantial returns not only in educational attainment and labor market outcomes but also in broader social and health outcomes.

Our estimates indicate that exposure to NAACP-movement-driven teacher-salary increases led to a meaningful improvement educational attainment in the short term,

socioeconomic status in the medium-term, and ultimately, longevity among Black students. This underscores the importance of addressing wage discrimination faced by Black teachers, as teacher salaries serve as a crucial indicator of schooling inputs and contribute significantly to school expenditure. By equalizing teacher salaries, we can address disparities in school resources, particularly in poorer districts, where limited resources may hinder educational opportunities for marginalized communities.

Moreover, our study sheds light on the long-term implications of equity-focused school finance reforms called *equity era* and *adequacy era* initiated since the 1970s. While previous research has highlighted the positive impacts of such reforms on short-term educational outcomes (Card & Payne, 2002; Jackson et al., 2016; Noghanibehambari, 2022; Rothstein & Schanzenbach, 2022), our findings provide valuable insights into their effects on longer-term outcomes, such as longevity.

Overall, our study highlights the importance of addressing structural inequalities in education and investing in policies that promote equity and equality. By prioritizing initiatives such as equalizing teacher salaries and implementing equitable school finance reforms, we can pave the way for a more just and equitable society, where all individuals have the opportunity to thrive and enjoy improved health outcomes.

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FIGURE06.JPEG

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6. Figures

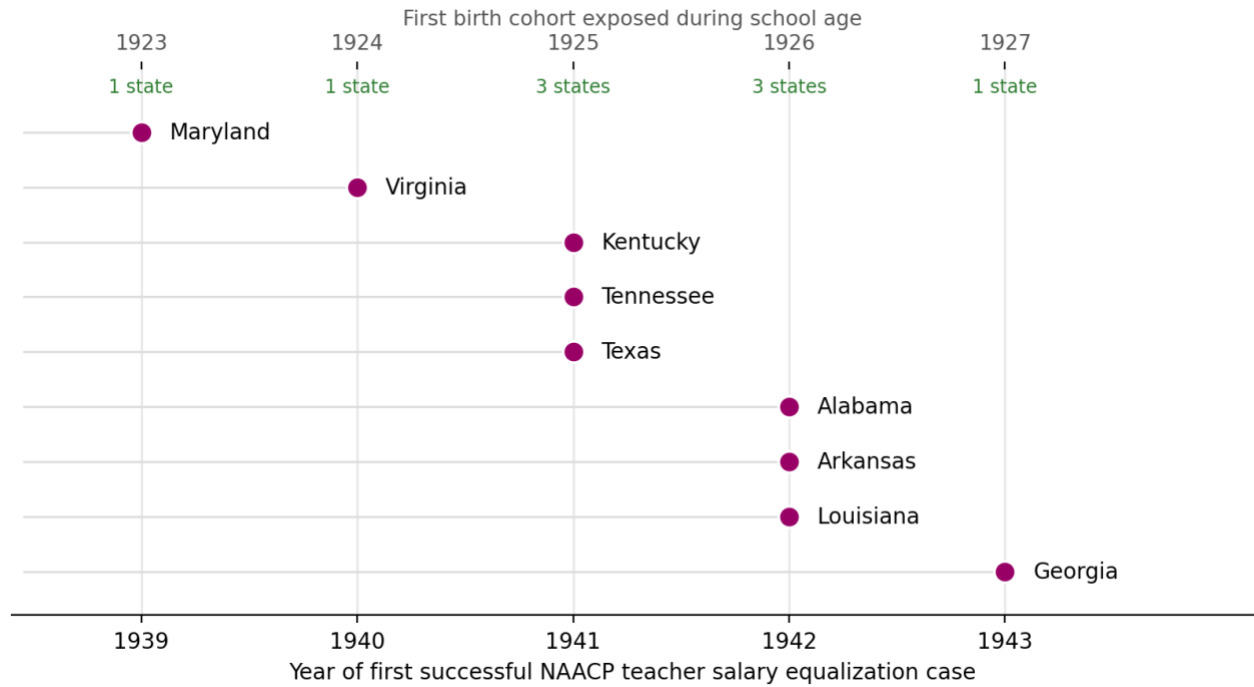


Figure 1 - NAACP-Induced Policies in the Final Sample

Notes. The figure presents the year of the first successful NAACP litigation in a federal court for each state in the final sample. Case timing is obtained from Murray (1947) and Cascio and Lewis (2024). The upper axis reports the first birth cohort exposed to equalization during school age, defined as the earliest cohort that had not yet turned 17 at the time of the case.

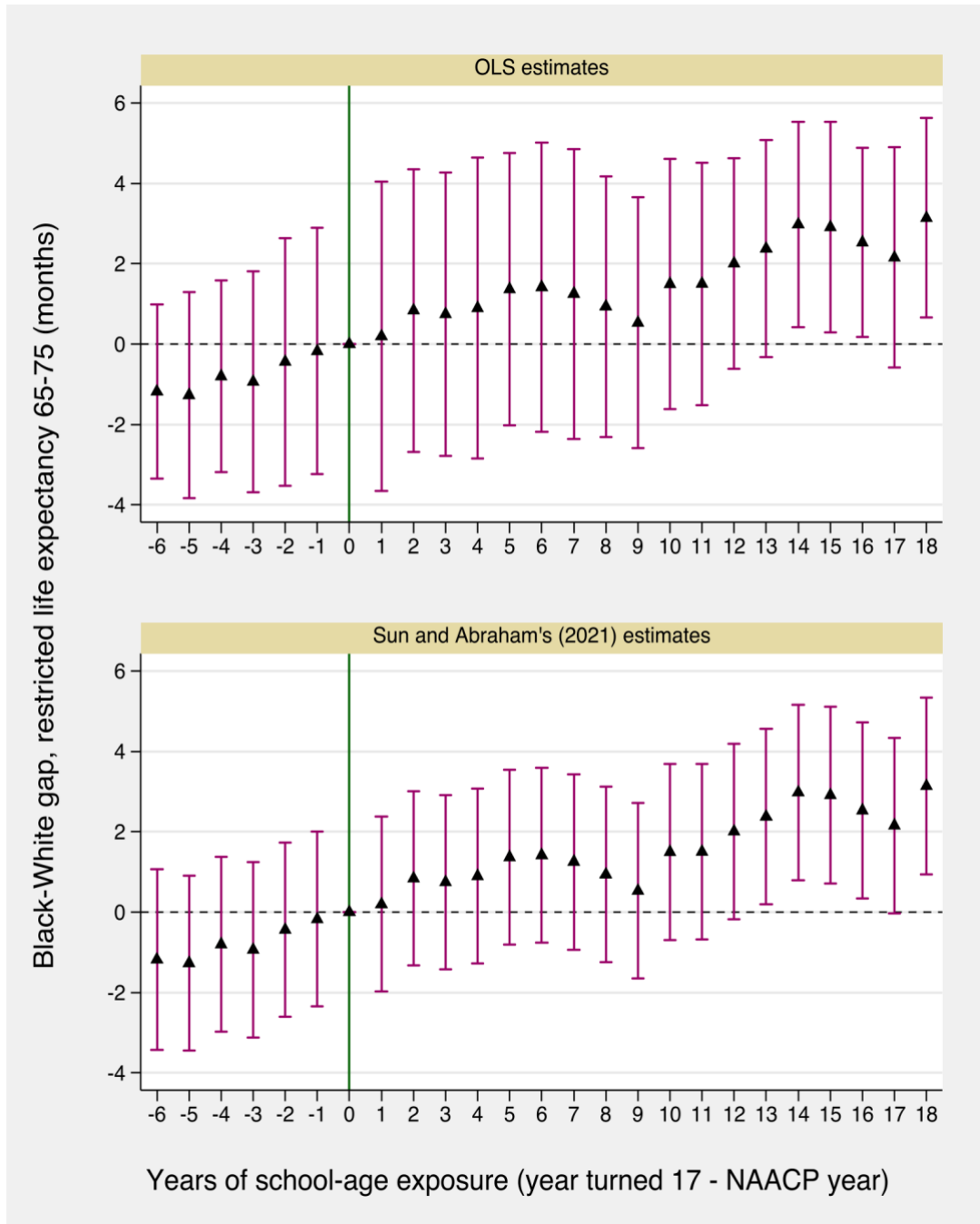


Figure 2 - Event Study Analysis

Notes. The vertical axis represents estimated effects in months, while the horizontal axis indicates the difference between the year an individual turned 17 and the year of the first successful NAACP litigation. Negative values in x-axis correspond to individuals who were older than 17 at the time of exposure, whereas positive values indicate those who were younger. The omitted (reference) cohort includes individuals who turned 17 in the year of the policy change, those who “just missed” being affected by it. Point estimates and 95% confidence intervals are reported. Standard errors are clustered on birth state and birth year. All regressions include birth state by birth year fixed effects, race and gender controls, and birth-state-level covariates. Birth state controls include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score. Data are from restricted-access death records from the National Center for Health Statistics for cohorts born between 1915 and 1945 who died between 1979 and 2020.

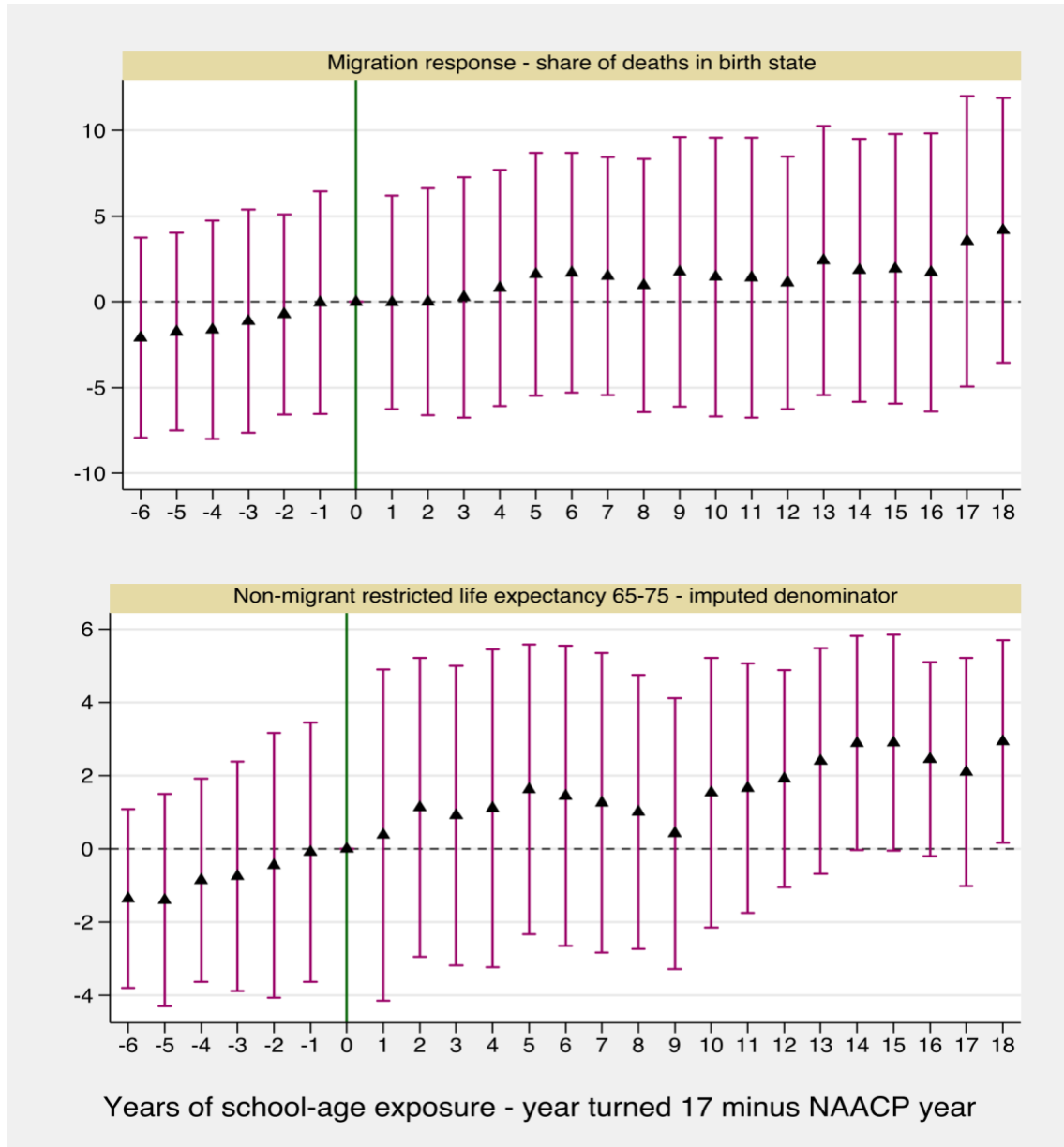


Figure 3 - Event Study Analysis – Non-Movers Sample

Notes. The figure reports event-study estimates from the non-migrant analysis. The top panel reports the share of deaths at ages 65 to 75 that occur in the individual's state of birth, expressed in percentage points and weighted by the number of deaths in the cell. The bottom panel reports restricted life expectancy between ages 65 and 75 among non-migrants, expressed in months, where the denominator is the birth-state census population scaled by the non-migrant share among decedents. The horizontal axis indicates the difference between the year an individual turned 17 and the year of the first successful NAACP litigation. Negative values correspond to individuals who were older than 17 at the time of exposure, whereas positive values indicate those who were younger and therefore had more years of school-age exposure. The omitted reference cohort comprises individuals who turned 17 in the year of the policy change, those who "just missed" being affected by it. Point estimates and 95 percent confidence intervals are reported. Observations are birth-state by birth-year by race cells. All regressions include birth-state by birth-year fixed effects and controls for race and the female share of the cell. Standard errors are clustered at the birth-state by birth-year cell level.

7. Tables

Table 1- Summary Statistics

	All cells	Black cells	White cells
<i>Panel A. Means weighted by the population alive at 65</i>			
Restricted life expectancy 65-75 in months	104.70 (2.42)	102.23 (3.39)	105.41 (1.42)
Survived to age 70 in percent	88.90 (1.91)	86.83 (2.59)	89.50 (1.09)
Survived to age 75 in percent	75.35 (3.51)	72.24 (5.15)	76.24 (2.17)
Exposed to NAACP equalization	0.72 (0.45)	0.73 (0.45)	0.72 (0.45)
Years of school-age exposure	6.08 (7.30)	6.29 (7.30)	6.02 (7.30)
Birth year	1930.40 (7.37)	1930.84 (7.38)	1930.28 (7.37)
<i>Panel B. Unweighted counts</i>			
Number of cells	450	225	225
Number of birth states	9	9	9
Number of birth cohorts	29	29	29

Notes. Observations are birth-state by birth-year by race cells. Restricted life expectancy 65–75 is the expected number of months of life lived between ages 65 and 75 among those alive at age 65. Deaths are from restricted-access records of the National Center for Health Statistics; the population alive at age 65 is constructed from decennial census counts by birth state, birth year, and race, interpolated to intercensal years. Panel A reports means weighted by the population alive at 65, with standard deviations in parentheses; Panel B reports unweighted counts. The sample comprises the 450 cells used in the main specification, covering birth cohorts of 1915–1945 born in nine Southern states with at least one successful NAACP Teacher Salary Equalization case by 1948, and deaths observed between 1980 and 2019.

Table 2– School Age Exposure to the Aftermath of the NAACP's First Successful Case and Later-Life Longevity

	<i>Outcome: Restricted life expectancy between ages 65 and 75 (months of life lived in that interval among those alive at 65)</i>		
	(1)	(2)	(3)
Exposure × Black	2.208*** (.461)	2.274*** (.466)	2.224*** (.469)
Exposure	-.482** (.216)	-.637*** (.213)	
Black	-4.582*** (.549)	-4.84*** (.559)	-4.767*** (.759)
Observations	450	450	450
R-squared	.775	.786	.81
Mean DV	104.703	104.703	104.703
Birth Year FE	✓	✓	✓
Birth State FE	✓	✓	✓
Race and Gender Controls	✓	✓	✓
Birth State Controls		✓	
Birth Year x Birth State FE			✓

Notes. Data are from restricted-access death records from the National Center for Health Statistics for cohorts born between 1915 and 1945 who died between 1979 and 2020. Standard errors, clustered at the birth state–birth year cell level, are reported in parentheses. Birth state controls include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score.

*** p<0.01, ** p<0.05, * p<0.1

Table 3 - Heterogeneity Analysis by Gender

	<i>Outcome: Restricted life expectancy between ages 65 and 75 (months of life lived in that interval among those alive at 65)</i>	
	Female	Male
	(1)	(2)
Exposure × Black	2.174*** (.351)	2.349*** (.591)
Black	-4.57*** (.276)	-5.984*** (.466)
Observations	450	450
R-squared	.78	.832
Mean DV	108.033	100.730
Birth Year x Birth State FE	✓	✓
Race and Gender Controls	✓	✓
Birth State Controls	✓	✓

Notes. Data are from restricted-access death records from the National Center for Health Statistics for cohorts born between 1915 and 1945 who died between 1979 and 2020. Standard errors, clustered at the birth state–birth year cell level, are reported in parentheses. All regressions include birth state by birth year fixed effects, race and gender controls, and birth-state-level covariates. Birth state controls include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score.

*** p<0.01, ** p<0.05, * p<0.1

Table 4 – Robustness Checks

	<i>Outcome: Restricted life expectancy between ages 65 and 75 (months of life lived in that interval among those alive at 65)</i>						
	Column 3 of Table 2	Adding birth state by race FE	Restricted to statewide salary- schedule states (AL, LA, TN, VA)	Outcome: log restricted life expectancy 65- 75	Outcome: P(survive to 70), in percentage points	Outcome: P(survive to 75), in percentage points	Clustering std errors on birth state
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Exposure × Black	2.224*** (.469)	1.938*** (.217)	1.968*** (.254)	.022*** (.005)	1.671*** (.352)	3.374*** (.727)	2.224*** (.474)
Observations	450	450	200	450	450	450	450
R-squared	.81	.946	.948	.809	.826	.784	.81
Mean DV	104.703	104.703	104.245	4.651	88.902	75.352	104.703
Birth Year FE	✓	✓	✓	✓	✓	✓	✓
Birth State FE	✓	✓	✓	✓	✓	✓	✓
Birth Year x Birth State FE	✓	✓	✓	✓	✓	✓	✓
Race and Gender Controls	✓	✓	✓	✓	✓	✓	✓

Notes. Data are from restricted-access death records from the National Center for Health Statistics for cohorts born between 1915 and 1945 who died between 1979 and 2020. Standard errors, clustered at the birth state–birth year cell level (except in column 7), are reported in parentheses. All regressions include birth year fixed effects, birth state fixed effects, birth year by birth state fixed effects, and race and gender controls.

*** p<0.01, ** p<0.05, * p<0.1

Table 5 – Exploring Mechanism

Outcomes:	Education = 0 years	High School Education	Occupational Prestige Score
	(1)	(2)	(3)
<i>Panel A. Women</i>			
Exposure × Black	-.014*** (.002)	.064*** (.007)	4.452*** (.36)
Black	.02*** (.002)	-.192*** (.005)	-17.982*** (.262)
Observations	558	558	558
R-squared	.97	.968	.97
Mean DV	0.027	0.561	38.164
Mean for Blacks	0.030	0.441	27.038
<i>Panel B. Men</i>			
Exposure × Black	-.019*** (.002)	.045*** (.008)	2.261*** (.266)
Black	.027*** (.002)	-.209*** (.005)	-16.82*** (.201)
Observations	558	558	558
R-squared	.975	.968	.981
Mean DV	0.030	0.552	36.712
Mean for Blacks	0.038	0.398	24.740

Notes. Data are from the decennial U.S. Census for 1950–2000, covering birth cohorts from 1915 to 1945. The sample includes only individuals who were aged 25 or older at the time of the Census in order to observe their completed high school education. Standard errors, clustered at the birth state–birth year cell level, are reported in parentheses. All regressions include birth state by birth year fixed effects, race and gender controls, and birth-state-level covariates. Birth state controls include share female in the birth state, literacy rate, average socioeconomic index, and average occupational income score.

*** p<0.01, ** p<0.05, * p<0.1

Table 6– Heterogeneity Analysis by Cause of Death

Outcome: Log Cumulative Mortality Rate 1979 – 2020 by Cause of Death:

	All Causes	Infectious Diseases (TB, Syphilis, Influenza, Pneumonia, other Infectious Diseases)	Chronic/Non-communicable Diseases (Neoplasm Diseases, Diabetes, Cardiovascular Diseases, Peptic Ulcer, Nephritis)	All other Diseases	External and Self- Inflicted Causes
	(1)	(2)	(3)	(4)	(5)
Exposure × Black	-.067*** (.022)	-.145*** (.027)	-.049** (.021)	-.08*** (.03)	-.258*** (.033)
Exposure	.018 (.017)	.054* (.026)	.023 (.015)	-.009 (.02)	.025 (.024)
Black	.198*** (.021)	-.042* (.024)	.239*** (.021)	.147*** (.029)	-.008 (.029)
Observations	558	558	558	558	558
Mean DV	10.189	8.145	9.840	8.144	6.648

Notes. The table presents heterogeneity analysis of log cumulative mortality rates (1979–2020) by cause of death between ages 65–75. Regressions are weighted using population counts in each cell. All regressions include birth state fixed effects and birth year fixed effects. Data are from restricted-access death records from the National Center for Health Statistics for cohorts born between 1915 and 1945 who died between 1979 and 2020. Standard errors, clustered at the birth state–birth year cell level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 7– Pre-Reform Teacher Salary Gaps by State

State	Black Teacher Annual Salary	White Teacher Annual Salary	Proportion Gap	Dollars Gap	Pupil/teacher	Dollars Gap/Black Pupil	Dollars Gap (\$2025)/Black Pupil
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Alabama	360	768	0.531	408	44.0	9.3	211
Arkansas	342	585	0.417	244	45.3	5.4	122
Georgia	317	793	0.600	476	41.9	11.4	259
Kentucky	607	802	0.243	195	34.1	5.7	130
Louisiana	451	1048	0.569	597	45.8	13.2	299
Maryland	1248	1569	0.205	322	35.4	9.1	206
Tennessee	561	868	0.355	307	38.1	8.0	183
Texas	630	1028	0.388	399	36.7	10.9	247
Virginia	539	963	0.439	424	37.9	11.2	255
Mean	561.5	935.9	0.416	374.4	39.9	9.4	212

Notes. The table reports pre-reform school inputs and the implied intensity of teacher salary equalization for the nine states in the estimation sample. School data are from Card & Krueger (1992b). Because the series is biennial, we use the school years 1936 and 1938, the last two observations preceding the first successful NAACP teacher salary equalization case in our sample (Maryland, 1939), and average across them. The proportional gap is (White – Black)/White. The dollar gap is White – Black. The Black pupil–teacher ratio is the number of Black pupils per Black teacher. Column (7) is the dollar gap divided by the Black pupil–teacher ratio, and measures the implied increase in annual teacher salary spending per Black pupil under full equalization. Column (8) deflates this quantity using the consumer price index, a factor of 22.7 relative to 1936–1938.

8. Appendix A. Simulation Evidence on the Choice of Mortality Measure

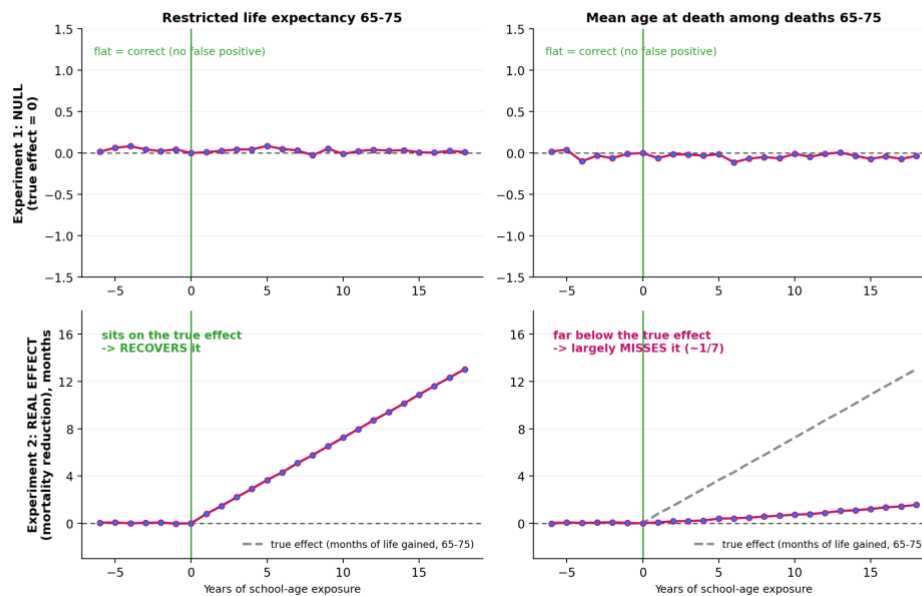
This appendix presents a Monte Carlo exercise that motivates the choice of mortality measure used in the paper. The simulation is intended to be illustrative rather than a model calibrated to the NCHS data. The results show that, within a balanced age window of 65 to 75, restricted life expectancy correctly captures a genuine reduction in mortality, whereas the mean age at death among individuals who die within the window does not. The latter measure is affected by selection because whether an individual dies inside the window is itself an outcome of the reform. When mortality falls, some individuals who would have died between ages 65 and 75 instead survive past 75. They do not die later within the window, they leave it altogether.

Whether this produces a visible bias depends on which deaths are averted. If the reform postpones deaths from across the whole window rather than only from its upper end, then the age distribution of the deaths that remain closely resembles the distribution that would have prevailed without the reform, and the conditional mean barely moves. This is what the lower row of Figure A-1 shows: the mean age at death recovers only about one seventh of the true gain.

Restricted life expectancy avoids the problem because its denominator is fixed by construction. It averages the years lived between 65 and 75 over everyone alive at 65, a population determined before the reform can act on it. Averted deaths are not dropped from the calculation; they appear as additional years of life lived inside the window. The measure therefore counts precisely what the mean age at death discards.

Figure A-1 - Restricted life expectancy versus mean age at death within the balanced band

Why restricted life expectancy 65-75, not mean age at death: both pass the NULL, but only the restricted measure detects a real mortality reduction



Balanced births 1915-1945 (no truncation artifact for either). Top: no true effect -> both flat. Bottom: a real mortality reduction (as the census CMR shows) -> restricted life expectancy rises with exposure; within-band mean age at death stays flat because the 'saved' deaths just leave the 65-75 band (collider). Matches the real data (restricted life expectancy 65-75 significant, mean death age flat). Under a hypothetical uniform lifespan shift the two would look similar; they diverge for this mechanism. All quantities are in months. Illustrative old-age-mode distribution.

Notes. Births 1915 to 1945 with a balanced 65 to 75 band. The top row imposes no true effect: both the restricted life expectancy and the mean age at death among those who die between 65 and 75 are correctly flat, so neither produces a false positive. The bottom row imposes a real mortality reduction of the kind implied by the census cumulative mortality rates. The restricted life expectancy rises with exposure and tracks the true effect, whereas the mean age at death remains nearly flat and recovers only about one seventh of the true gain. The reason is that averted deaths leave the 65 to 75 window entirely, so conditioning on dying inside the window conditions on an outcome of the treatment. All quantities are in months.